

A Left-Universal Operator on $\mathcal{C}([0, 1])$

Candidate full solution to a question in arXiv:1912.13312

Source and Open Question

The source paper is Joanna Garbulinska-Wegrzyn and Wieslaw Kubis, *A note on universal operators between separable Banach spaces*, arXiv:1912.13312 [1]. After proving that there exists a universal operator on $\mathcal{C}([0, 1])$, the authors write:

We do not know whether there exists a left-universal operator on $\mathcal{C}([0, 1])$.

The passage appears on page 4 of the source PDF:

0171.

□

As a consequence, there exists a universal operator on $\mathcal{C}([0, 1])$. We do not know whether there exists a left-universal operator on $\mathcal{C}([0, 1])$. The situation changes when replacing $[0, 1]$ with the Cantor set $2^{\mathbb{N}}$. The space $\mathcal{C}(2^{\mathbb{N}})$ is linearly isomorphic (but not isometric) to $\mathcal{C}([0, 1])$ and it is isometrically universal, too. Furthermore, $\mathcal{C}(2^{\mathbb{N}}) \oplus \mathcal{C}(2^{\mathbb{N}})$ with the maximum norm is linearly isometric to $\mathcal{C}(2^{\mathbb{N}})$, because the disjoint sum of two copies of the Cantor set is homeomorphic to the Cantor set. Thus, Example 1.2 provides a left-universal operator on $\mathcal{C}(2^{\mathbb{N}})$.

Another, not so well known, universal Banach space is the *Gurarii space*. This

Definitions

Following the source paper, a bounded linear operator $U : V \rightarrow W$ is *left-universal* for operators into W if for every bounded linear operator $T : X \rightarrow W$, with X separable and $\|T\| \leq \|U\|$, there is a linear isometric embedding $i : X \rightarrow V$ such that

$$Ui = T.$$

Proof Intuition

The projection $\mathcal{C}([0, 1]) \oplus_{\infty} \mathcal{C}([0, 1]) \rightarrow \mathcal{C}([0, 1])$ is left-universal because the first coordinate can carry an isometric copy of the domain and the second coordinate can carry the given operator. The point is that this two-coordinate argument can be placed inside one copy of $\mathcal{C}([0, 1])$. Use the left third of the interval as the first coordinate, the right third as the second coordinate, and connect the endpoint values over the middle third. The middle bridge is a convex interpolation, so it never increases the supremum norm.

Theorem 1. *There exists a norm-one left-universal operator*

$$U : \mathcal{C}([0, 1]) \longrightarrow \mathcal{C}([0, 1]).$$

In particular, the question from [1] has an affirmative answer.

Proof. Define

$$(Uf)(t) = f\left(\frac{t+2}{3}\right), \quad f \in \mathcal{C}([0, 1]), \quad t \in [0, 1].$$

Clearly $\|U\| \leq 1$, and equality holds by applying U to the constant function 1.

Let $T : X \rightarrow \mathcal{C}([0, 1])$ be a bounded linear operator, where X is separable and $\|T\| \leq 1$. Since $\mathcal{C}([0, 1])$ is isometrically universal for separable Banach spaces, choose a linear isometric embedding

$$e : X \longrightarrow \mathcal{C}([0, 1]).$$

Define $I : X \rightarrow \mathcal{C}([0, 1])$ by the following formula. For $x \in X$ and $s \in [0, 1]$, set

$$(Ix)(s) = \begin{cases} e(x)(3s), & 0 \leq s \leq \frac{1}{3}, \\ (2 - 3s)e(x)(1) + (3s - 1)T(x)(0), & \frac{1}{3} \leq s \leq \frac{2}{3}, \\ T(x)(3s - 2), & \frac{2}{3} \leq s \leq 1. \end{cases}$$

The endpoint values agree at $s = 1/3$ and $s = 2/3$, so Ix is continuous. The formula is linear in x .

We check that I is an isometric embedding. For each $x \in X$, the left third gives

$$\sup_{0 \leq s \leq 1/3} |(Ix)(s)| = \|e(x)\| = \|x\|.$$

On the right third,

$$\sup_{2/3 \leq s \leq 1} |(Ix)(s)| = \|T(x)\| \leq \|x\|.$$

On the middle third, $(Ix)(s)$ is a convex combination of the two endpoint scalars $e(x)(1)$ and $T(x)(0)$, so its modulus is at most $\max\{|e(x)(1)|, |T(x)(0)|\} \leq \|x\|$. Thus $\|Ix\| = \|x\|$ for every $x \in X$.

Finally, for every $t \in [0, 1]$,

$$(U(Ix))(t) = (Ix)\left(\frac{t+2}{3}\right) = T(x)(t),$$

because $(t+2)/3 \in [2/3, 1]$. Hence $UI = T$. This proves that U is left-universal. $\square$

Verification Notes

The proof is explicit and uses no computation. The only external input is the classical Banach–Mazur universality of $\mathcal{C}([0, 1])$ for separable Banach spaces, recalled in the source paper. The construction works over both real and complex scalars; in the complex case the middle-third estimate follows from the triangle inequality applied to a convex combination.

Novelty Check and Limitations

Local duplicate checks in the registry, solution, attempt, and proof-gap indexes found no previous packet for arXiv:1912.13312 or for the left-universal $\mathcal{C}([0, 1])$ question. Web searches on 2026-06-19 for exact phrases including “left-universal operator”, “ $\mathcal{C}([0, 1])$ ”, and the source title surfaced the source paper but no later explicit answer. The packet should therefore be treated as a plausible new affirmative solution, subject to human literature review.

Human Review Recommendation

Review should focus on the piecewise definition of I , especially the norm calculation and the identity $UI = T$. If those two elementary checks pass, the construction gives a full affirmative answer.

References

- [1] J. Garbulinska-Wegrzyn and W. Kubis, *A note on universal operators between separable Banach spaces*, arXiv:1912.13312, 2019.